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Shefaa Smart Clinic Management System

A Proposal Presented for Fulfillment
of The Diploma Project in Computer Science

Submitted by

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Introduction

Due to the increasing pressure on healthcare services and the lack of efficient digital systems in many medical clinics, patients often face difficulties in booking appointments, long waiting times, and poor organization of medical records. Most clinics still rely on manual methods such as phone calls or paper-based schedules, which leads to appointment conflicts, data loss, and inefficient use of time.

As a result, we attempt to create a comprehensive smart web-based system to manage medical clinics digitally. The Sefaa Smart Clinic Management System aims not only to facilitate appointment booking for patients, but also to support doctors and clinic administrators in managing schedules, medical records, and clinic operations efficiently. This is accomplished through the implementation of the Shefaa system as a graduation project at the Institute of Statistical Studies and Research.

Project Background

With the rapid development of web technologies, digital transformation has become essential in the healthcare sector. While large hospitals often use advanced information systems, small and medium-sized clinics still lack integrated platforms for managing appointments and patient data. Many clinics depend on basic communication tools such as phone calls or messaging applications, which are inefficient and unorganized.

In this project, we extend the traditional clinic booking concept to build an integrated smart healthcare platform. The Shefaa system combines appointment scheduling with patient management, doctor management, electronic medical records, reporting, and notification services. This integration improves workflow efficiency and enhances the quality of healthcare services provided by clinics.

Problem Statement

Patients face difficulties in booking clinic appointments easily, so they rely on phone calls or manual registration. This leads to appointment overlaps, long waiting times, and overcrowding inside clinics. Additionally, clinics suffer from poor management of patient records and lack of analytical tools to support decision-making.

The Shefaa Smart Clinic Management System allows patients to book appointments from one place and enables clinics to manage appointments, patient information, and medical records through a single unified web-based platform.

Goal

The main goal of this work is to develop an integrated smart healthcare platform that facilitates clinic appointment booking and supports efficient management of medical clinics and patient records.

Objectives

1. Provide an integrated smart platform for managing medical clinics.
2. Enable patients to book, cancel, and reschedule appointments online.
3. Assist doctors in managing daily schedules and patient medical records.
4. Create and maintain electronic medical records for patients.
5. Reduce waiting times and overcrowding inside clinics.
6. Generate analytical reports to support administrative decision-making.
7. Enhance communication through automated notifications and reminders.
8. Ensure data security through authentication and role-based access control.

Beneficiaries

The priority beneficiaries of the system are patients, doctors, clinic staff, clinic administrators, and healthcare organizations.

Proposed Methodology

1. Create the forms and pages of the system for different user roles.

2. Create the database files needed for clinics, doctors, patients, appointments, and medical records.
3. Manage the structure and architecture of the system.
4. Develop the backend and frontend code of the system.
5. Manipulate and organize system data and contents.
6. Apply security and authentication mechanisms.
7. Obtain the final version that contains system pages, database files, and documentation.

Languages and Libraries

- SQL Database.
- C# with ASP.NET Core Web API [1], [2].
- JavaScript (TypeScript) language and Angular framework [4], [5].

Tools and Frameworks

- SQL Server [3].
- ASP.NET Core Framework [1].
- Angular Framework [4].
- HTML [5].
- Cascading Style Sheets (CSS) [5].
- Bootstrap [6].
- Source control (GitHub) [7].
- MS Office Programs.

Project Plan

1. Gather healthcare system requirements and clinic workflow details.
2. Design the database and system architecture.
3. Design the system screens and user interfaces.
4. Build the database and backend services.
5. Implement frontend and backend code.
6. Test the project.

References

[1] https://learn.microsoft.com/aspnet
[2] https://learn.microsoft.com/dotnet
[3] https://learn.microsoft.com/sql
[4] https://angular.io
[5] https://www.w3schools.com
[6] https://getbootstrap.com
[7] https://github.com